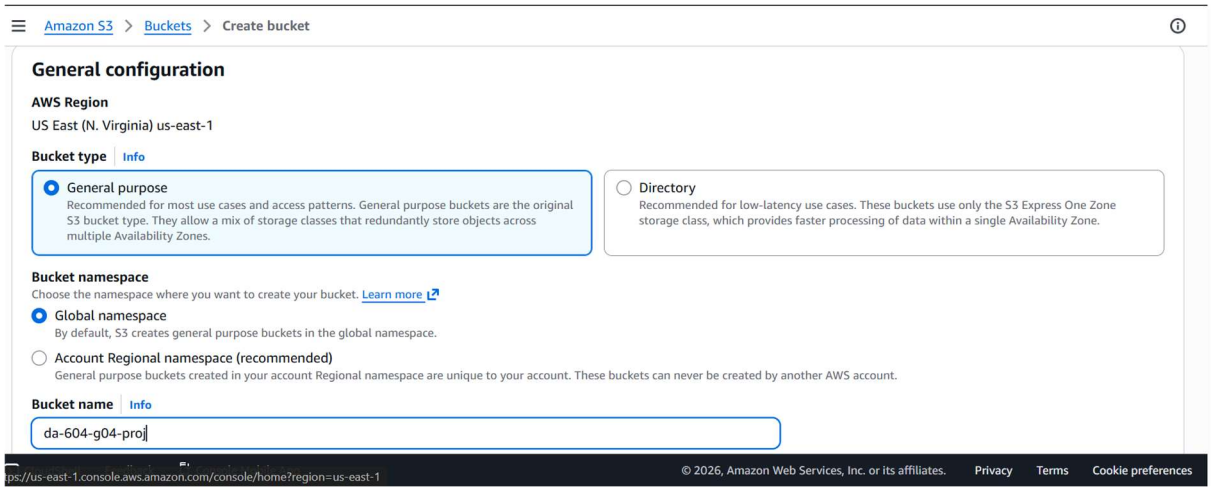
Guide to Configuring AWS Settings

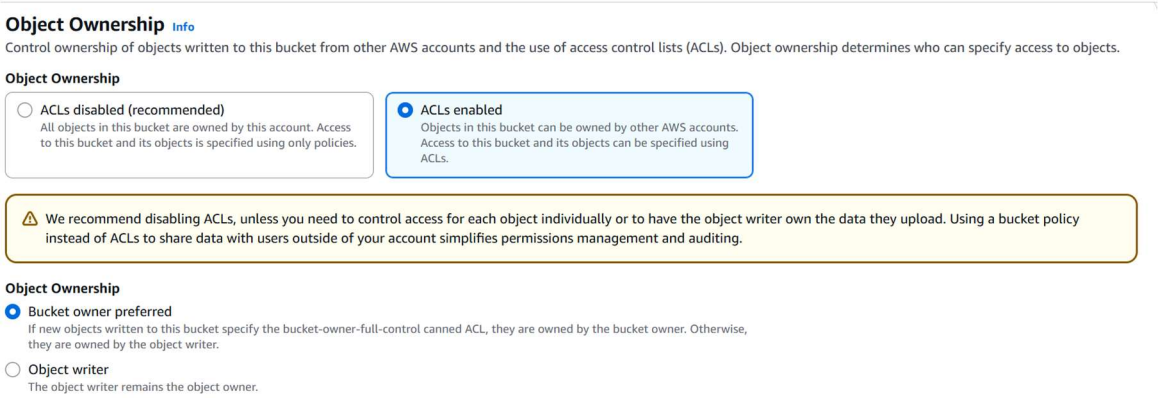
A guide to the Workflow Steps up to Creation of Combined Data Parquet Files

Step 1: Creation of S3 Bucket

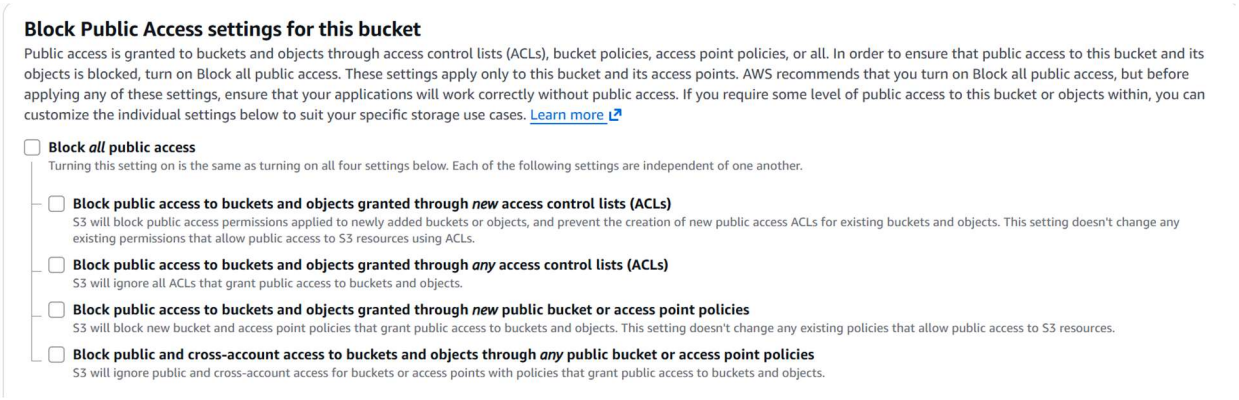
Create the bucket with general purpose bucket type and global namespace. Give the bucket a unique name.

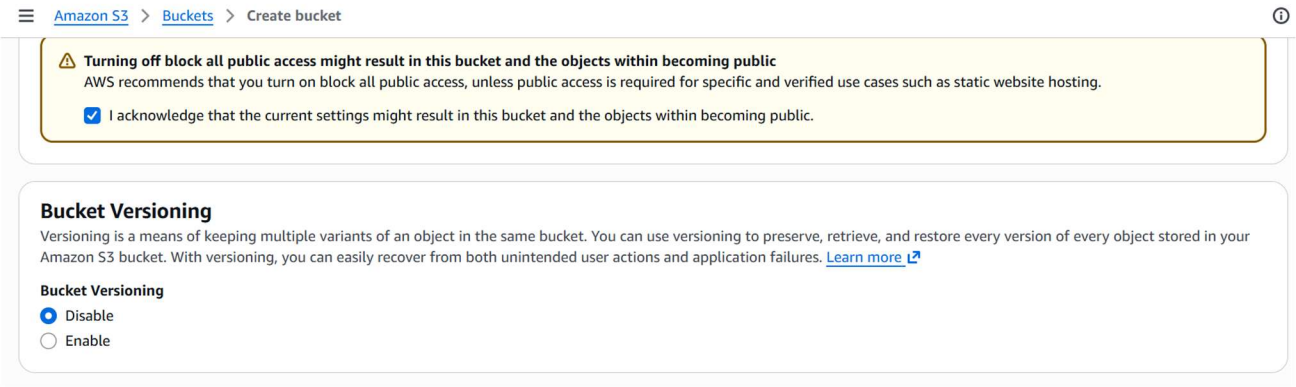


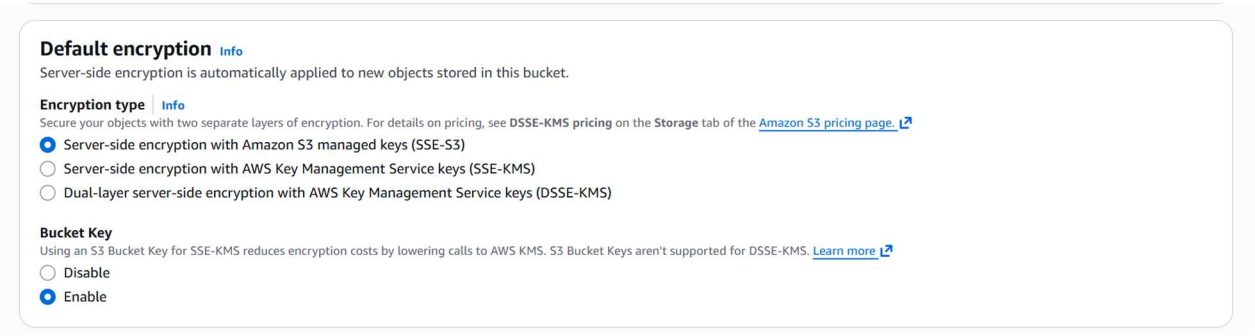
Enable ACLs – This is so we can later make the finalized parquet files and the airport coordinates list public.



Uncheck the block all public access – all the data used in this project is publicly available so we are not concerned with data privacy.

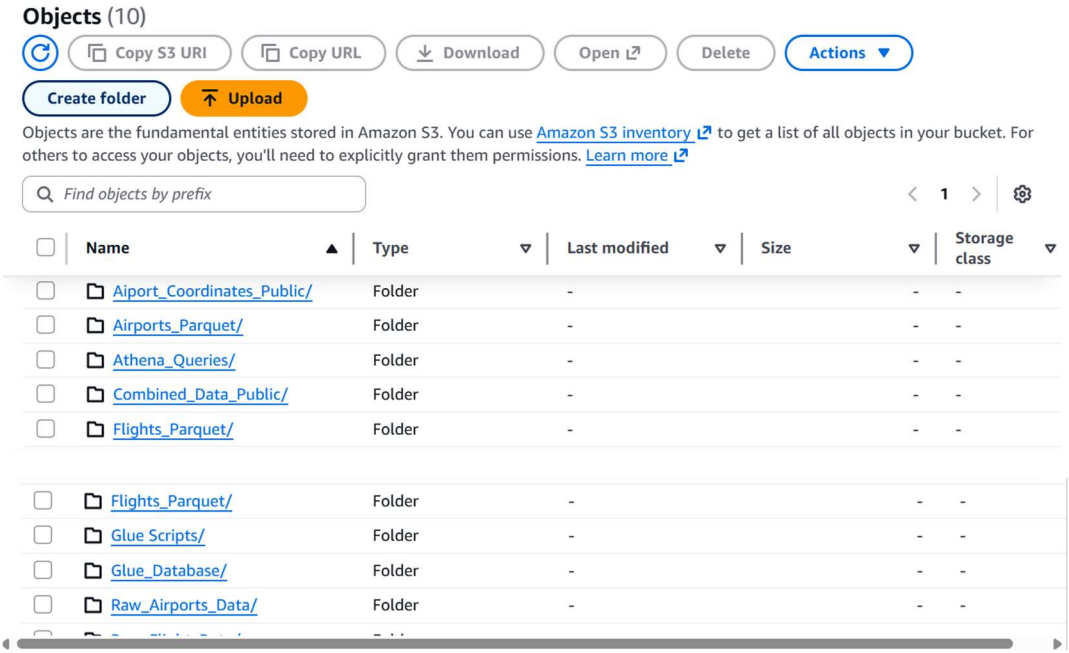






Step 2: Set up the folders on the S3 Bucket

Set up the following 10 folders in the S3 bucket. These will be needed to hold specific files. You can use “Create folder” button to do this.



Step 3: Upload the raw flight files by navigating to the Raw_Flight_Data and clicking on upload. Select Add files and you will be able to add the files from local computer. Once files are all listed, select the upload button at the bottom.

Step 4: Upload the airport listings csv from BTS portal to the Raw_Airports_Data folder using the same process as step 3.

Step 5: Optional Set up of role needed for glue.

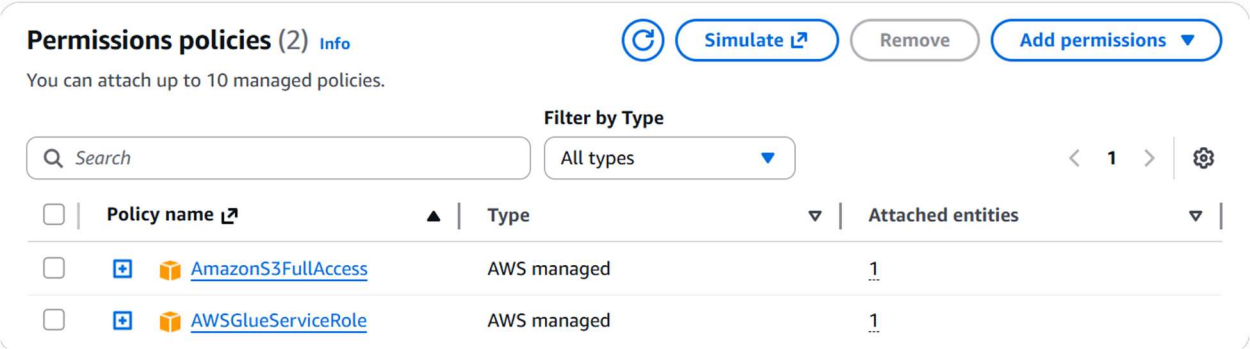
When using the AWS Learner Lab we use the lab role for glue, but when you use the AWS free tier you have to configure the role yourself.

To do this navigate to the IAM console.

Navigate to "Roles" and select "Create Role."

Under Step 1 of Create Role: select AWS Service for Trusted Entity Type. For the Use Case select Glue.

Under Step 2 Create Role: select the following permission policies



Under Step 3 of Create Role: give the role a name and description and click "Create Role." Do not adjust the Trust Policy script.

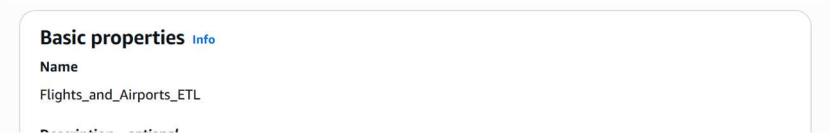
Step 6: Glue Data Catalog Data base set up.

A glue data catalog database has to be set up to hold the schema for the "tables" for amazon athena. Navigate to Databases under the data catalog and choose the default catalog. Select Add database. Create a Glue Database and give it an appropriate name and description.

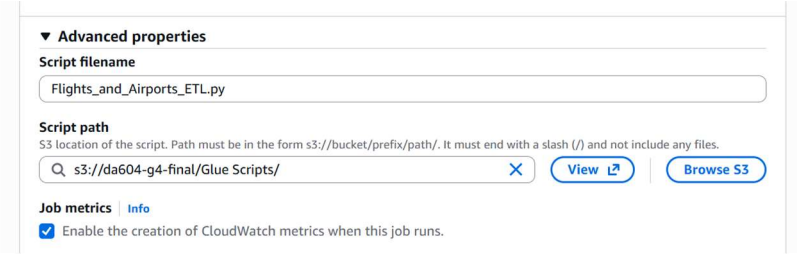
Step 7: Glue Visual ETL Set Up

1. Navigate to AWS Glue Studio.
2. Select Visual ETL
3. Under "Job Details" tab for IAM role, if you are using AWS Learner Lab select the LabRole, if you are using an AWS free tier or paid account select the role you created in Step 5. This is important, as it is required for Amazon Glue to talk to make make changes to your bucket.

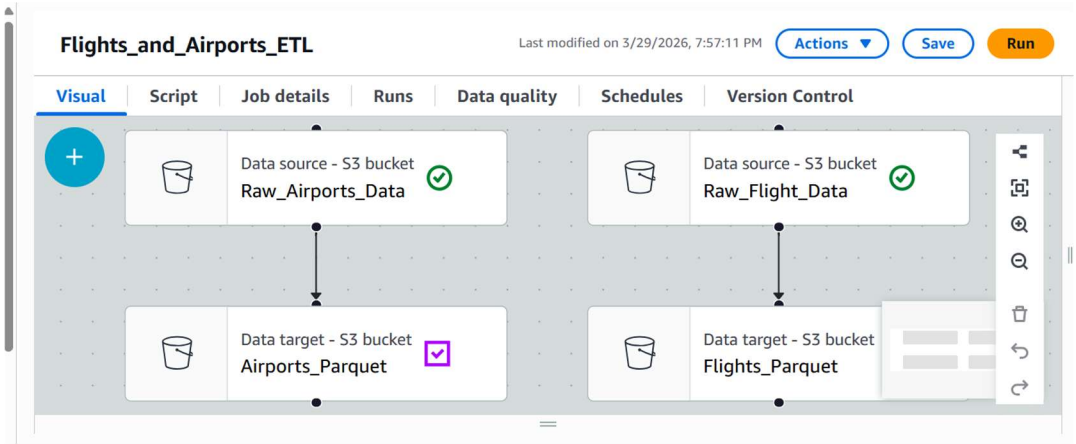
4. Under Job Details name the ETL something appropriate



5. Do not modify anything else under basic properties. Navigate to Advanced properties under the Job Details tab.
6. Under Advanced Properties change the script path to the path for your Glue_Scripts folder in S3.



7. Navigate back to the Visual tab and drag in the source data and the target data. Configure as follows:



For the Data sources, ensure data format of CSV is selected and path to raw csv files is specified. Check off box for “First line of source file contains column headers” and ensure Quote character is set to Double Quote (this is needed due to the nature of the strings in the csvs that have a comma between city and state).

For the output files select output file type as “Parquet” and compression type as “GZIP”.

Data Catalog update options: select “Create a table in the Data Catalog and on subsequent runs, update the schema and add new partitions”

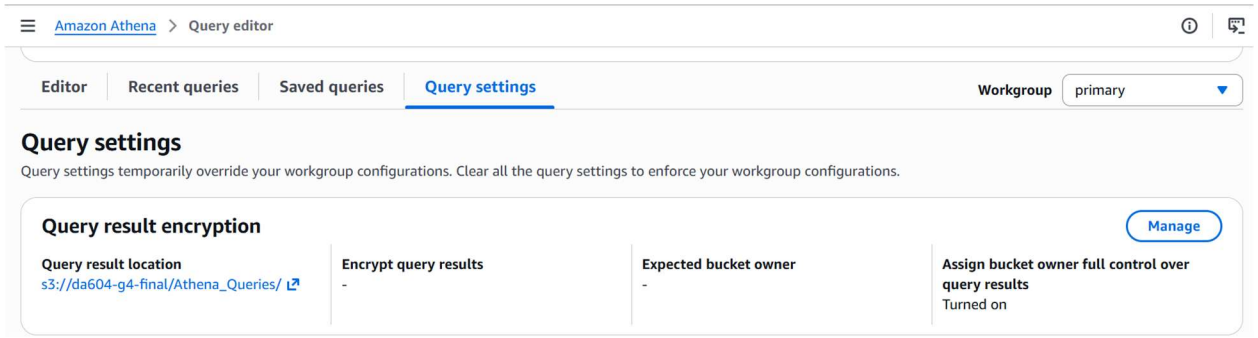
Under Database select the database you created in Step 6: Glue Data Catalog Data base set up. Name the table for the flight data flights. Name the table for the airport data airports.

When you are finished configuring the job. Save it and run it.

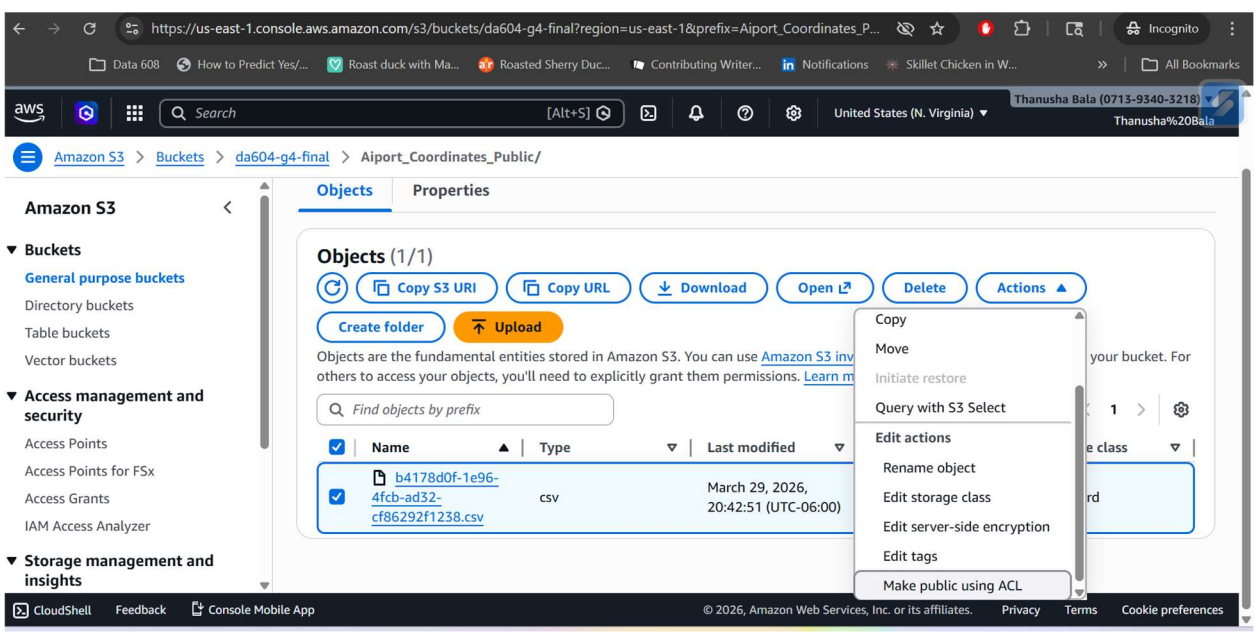
Click “run details” to see the status of the run.

Step 8: Amazon Athena Queries for initial checks and getting the unique airports. Please refer to github to get the queries ran as well as results.

1. Go to Amazon Athena
2. Go to Query your Data in Athena console and Launch the Query editor.
3. Under Query Settings change the query result location to the Athena_Queries folder created in your S3 bucket.

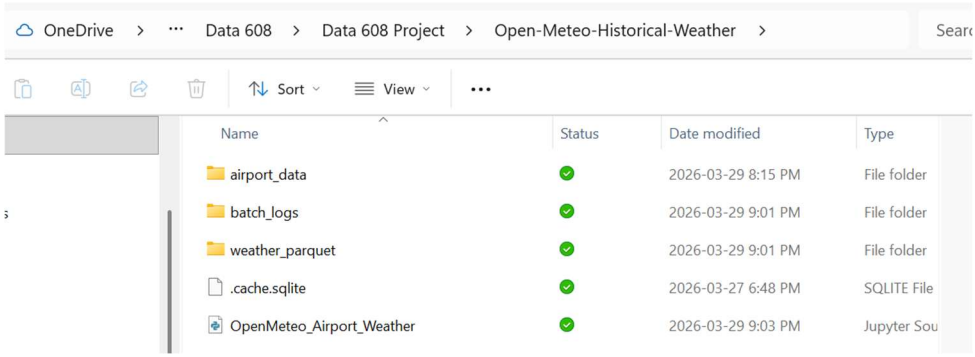


4. Run the Count_Total_Records query and the Count_Flights_By_Month query to get the total records, as well as the number of flight records for each month in the dataset. We will reconcile this to the csv files.
5. Run the Unique_Airports query. This will show how many unique airports are in our historical flight data and will list them.
6. Run the "Unique_Airport_Code_And_Coordinates" quer. This query builds upon the Unique_Airports query by adding in the latitude and longitude from the airports listing data to the unique airports as well as the utc_time_variation. We will need to export the csv of results.
7. Take the csv of results that was exported and save it in the Aiport_Coordinates_Public folder.
8. Navigate to the object in the S3 bucket check it off and under actions select, "Make Public Using ACL.



Step 9: Use the OpenMeteo_Airport_Weather python notebook and run it on local to get the parquet files for the airport weather. See github for link.

1. Set up a main folder to hold the python script and the subfolders in the following way:



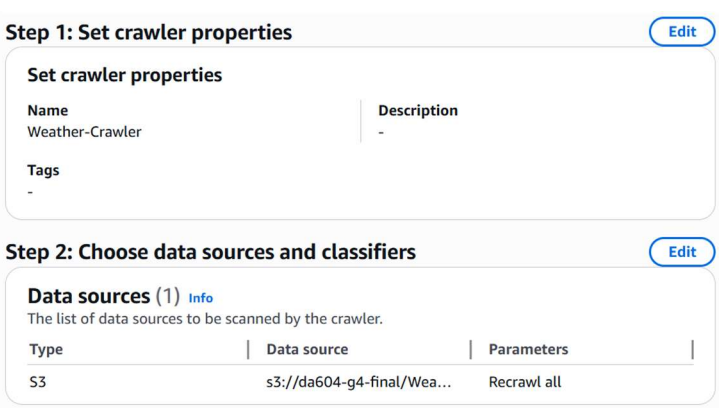
2. Refer to the OpenMeteo_Airport_Weather for instructions on how to adjust batch sizes and dates on the API calls.
3. Please note, if part of a batch fails, you can re-run the batch and it will only pull for failed airports.
4. You should have a single parquet file for each airport, you can cross reference with the airport coordinates csv from Step 8 to ensure completeness.

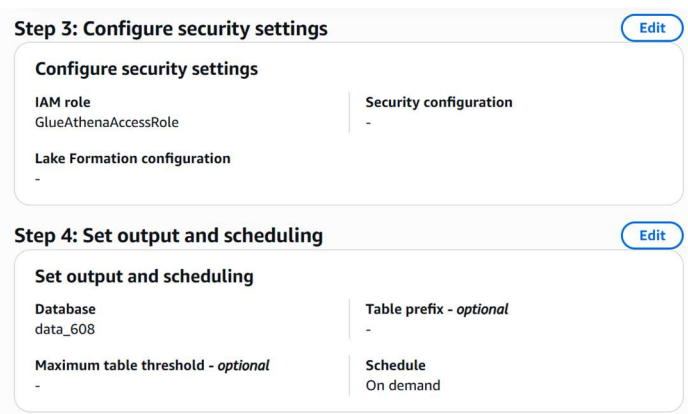
Step 10: Load the parquet files from Step 9 from local machine folder into S3 bucket in the Weather folder.

Step 11: Configure Glue Crawler to get schema for weather parquet files into data catalog.

Although the weather files are in parquet format, amazon athena will not be able to query it until we save the schema as a glue database table. To do this we will use a Glue Crawler. Unlike the Visual ETL, Glue Crawler does not transform data. Glue Crawler is just an automated way of getting the data.

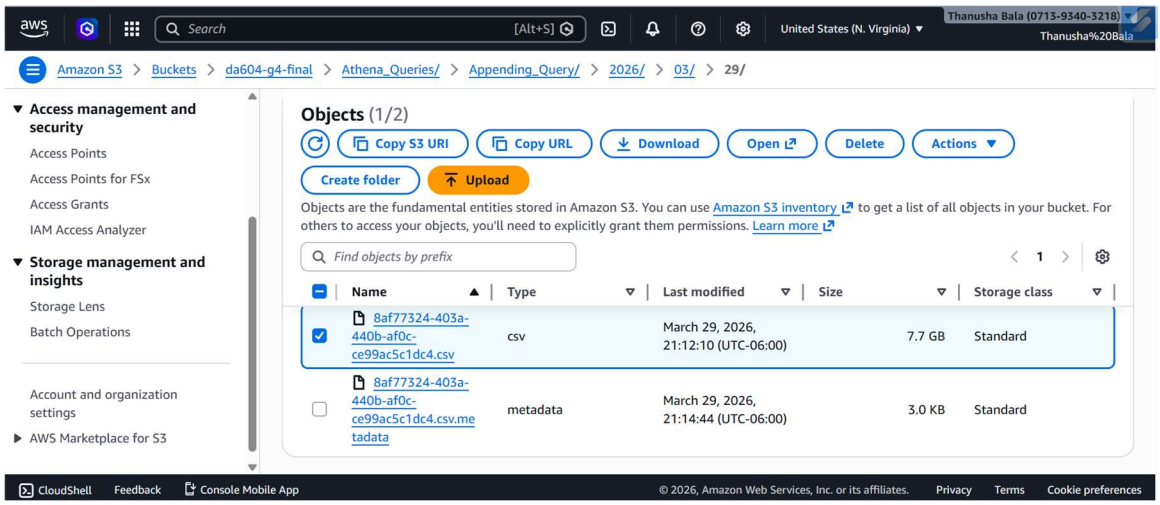
Follow Screenshots to set up crawler:





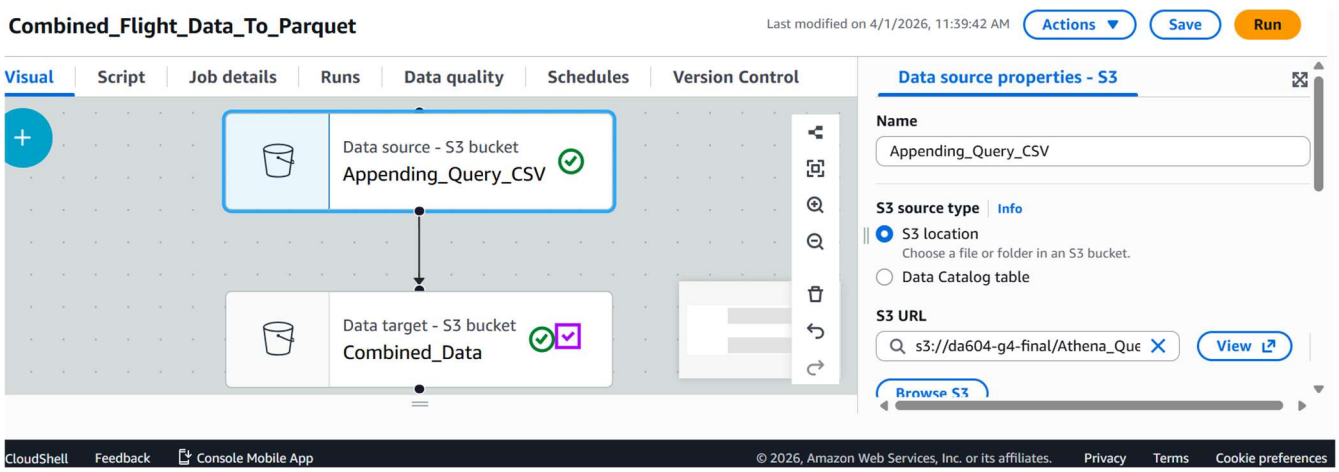
Step 12: Joining Weather to Flight Data (Refer to github for query)

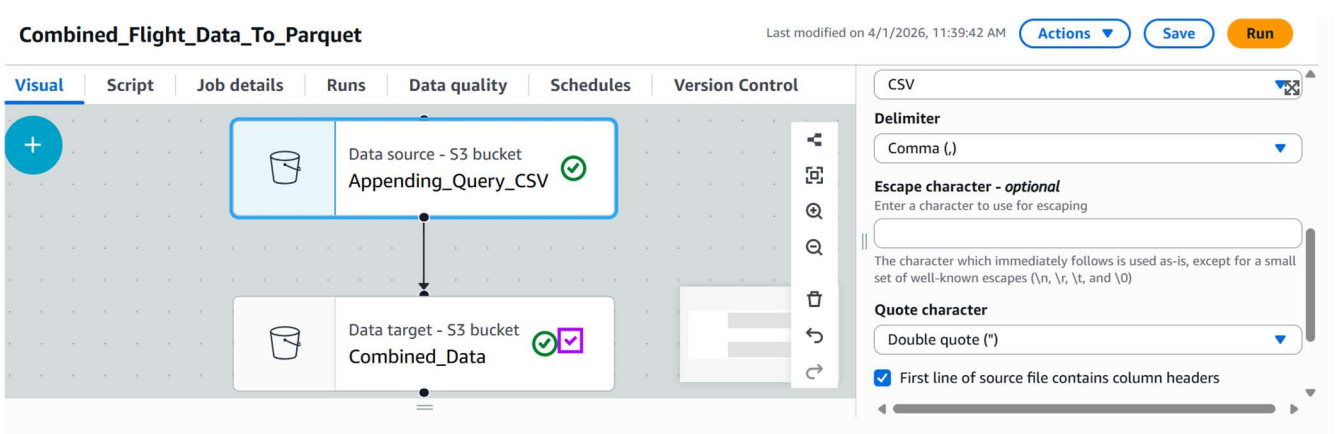
1. Run the Appending_Query, ensure the number of results matched the number of records in the flight data.
2. Navigate in the S3 Bucket to the Athena_Queries.
3. Got to Appending_Query. Find the location of where the query results csv was saved. Click on the csv and hit the "Copy s3 URI" button, you will need this for step 13.



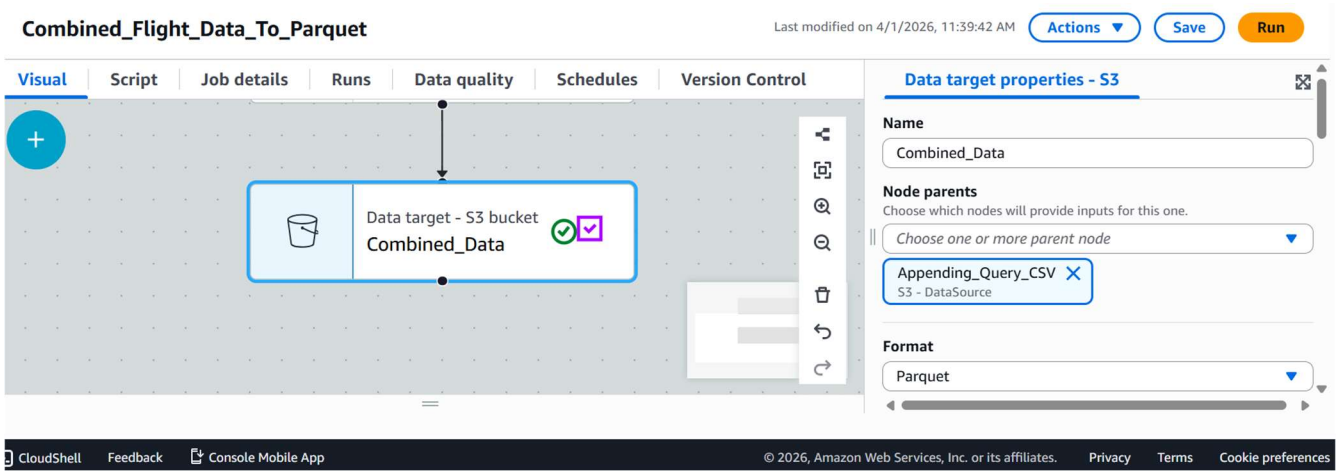
Step 13: Use Visual ETL to transform the csv of the appending query results into parquet files and create a combined data table.

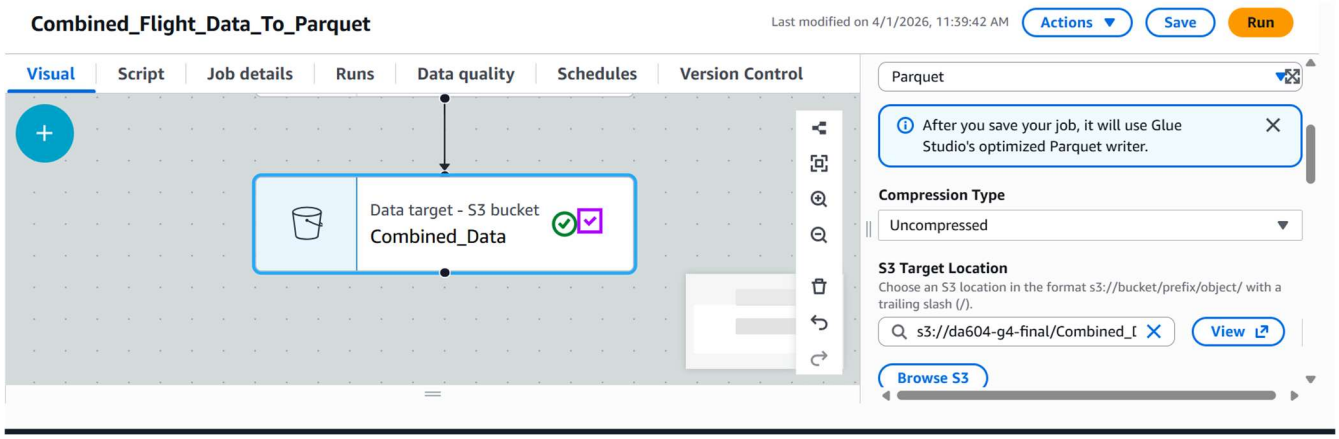
1. Open up a new tab and navigate the Glue Studio Visual ETL tool.
2. Configure the role under job details the way you did for the previous ETL.
3. Paste the S3 URI path you copied in Step 12 into the S3 URL for the Source node.

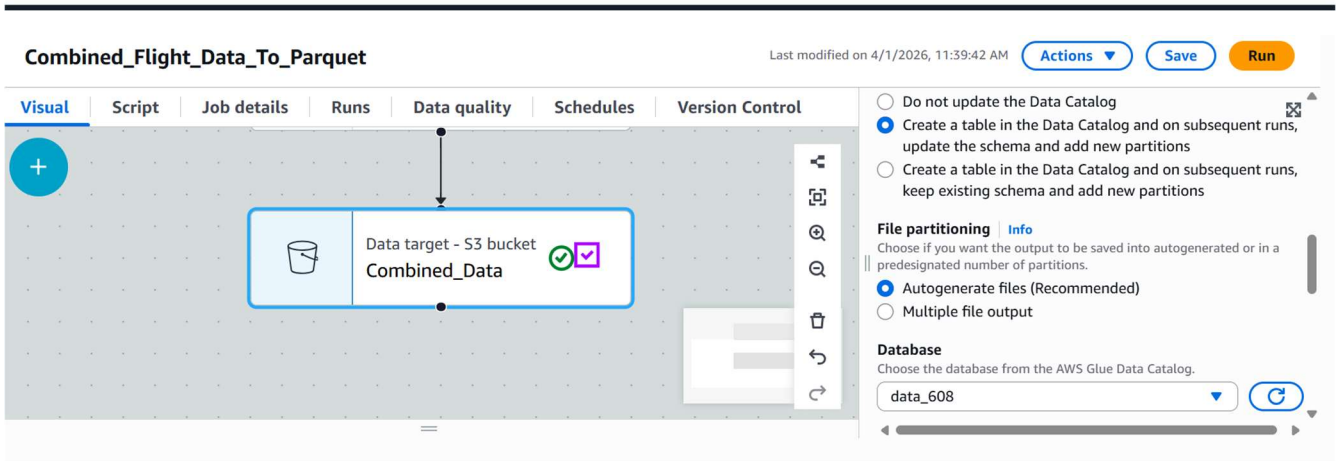


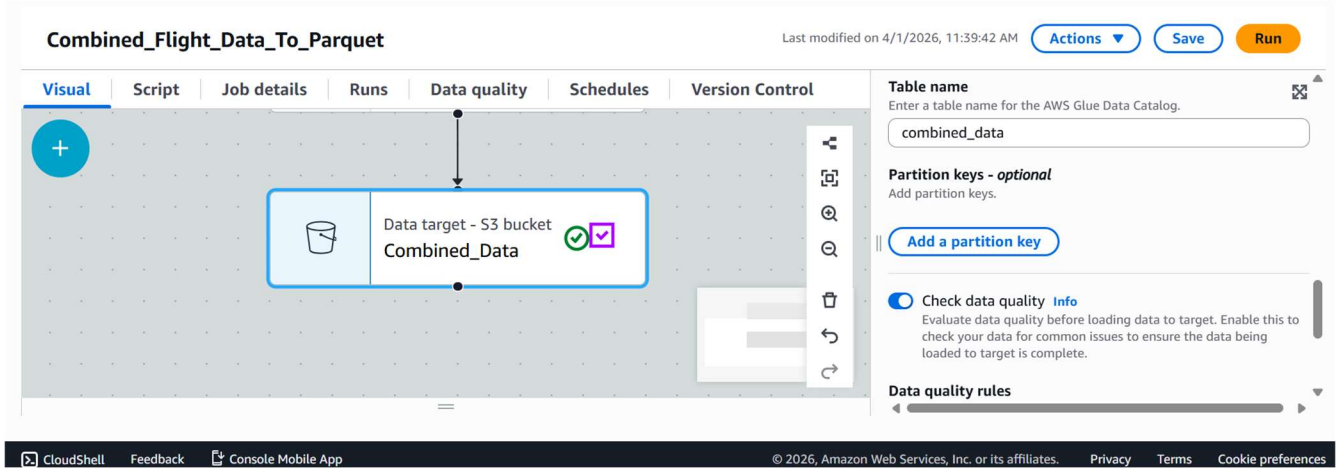


4. For the target node, select parquet files and uncompressed. Ensure the out put is saved in the Combined_Data_Public folder in the AS3 bucket.

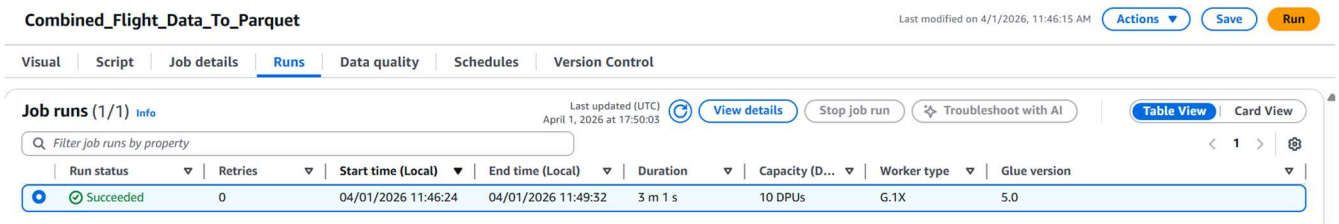








5. Save and run the Job.
6. Select run details to check status of run.



Step 14: Run the "Count_Flights_By_Month" on the combined_data table created in Step 13. Run additional spot checks as needed.

Step 15: Make the parquet files in the Combined_Data_Public folder public so it can be used for modeling on another person's instance. If the same account that owns the bucket will run the model on an EC instance the data can be kept private.